OpenTelemetry Web 101

Instrumentation of web apps

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About me

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// Biography

Passionate about OpenTelemetry, observability, and building scalable web applications. Formerly developer of web and mobile applications.

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Real User Monitoring (RUM) Players

Dynatrace	Datadog	New Relic	Splunk	Elastic
AppDynamics	Raygun	SpeedCurve	Catchpoint	IBM Instana

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What is OpenTelemetry?

Observability framework designed to generate, export and collect telemetry.

100% open-source and vendor agnostic.

Not an observability backend. You own the data and decide where it goes.

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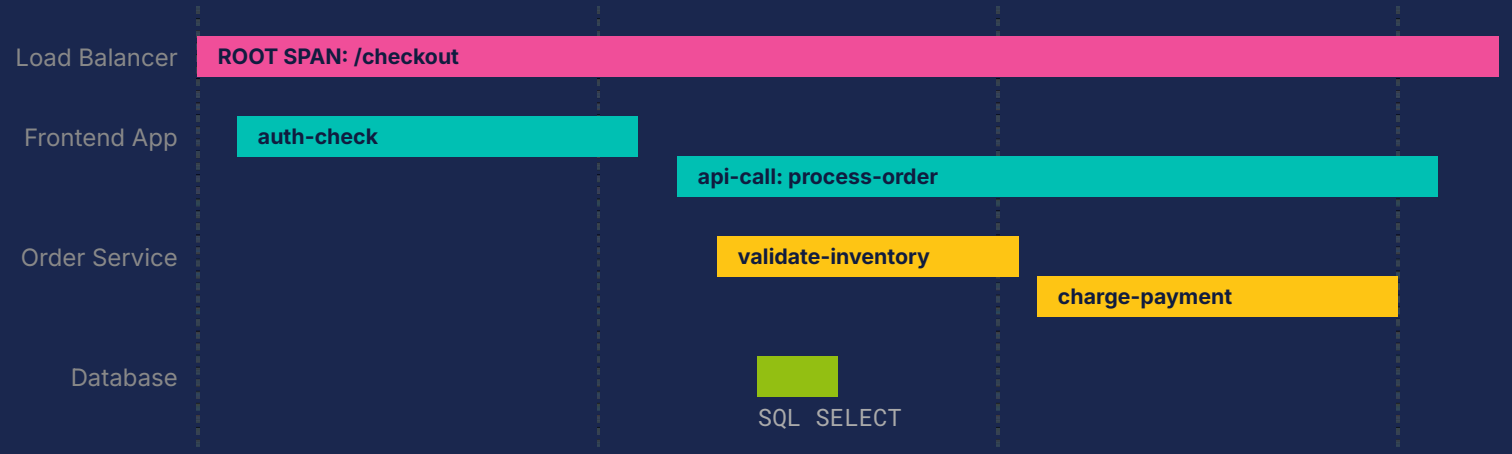
Tracing: Following the Journey of a Single Request

Traces provide a complete, end-to-end picture of how requests move through distributed systems, breaking down latency into individual spans.

```
// Standardized vendor-agnostic OTLP protocol
span.setAttribute("customer_id", "c-123");
```



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OpenTelemetry Signals: Metrics

Metrics are numerical measurements of service health and performance over time. Unlike traces, they provide aggregated visibility into system state and resource utilization.

Key Metric Types:

- **Counters:** Value that only increases (e.g., total requests, jobs processed).
- **Gauges:** Current instantaneous value (e.g., queue depth, memory usage).
- **Histograms:** Statistical distribution of values (e.g., latency percentiles).

```
// OTLP Gauge Example
meter.gaugeBuilder("queue.depth")
```

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OpenTelemetry Signals: Logs

Logs are discrete events or records that occur within a system. Unlike metrics or traces, logs provide high-cardinality, granular detail about specific moments in time.

Log Attributes:

- **Timestamp:** When the event occurred.
- **Severity:** Importance level (INFO, WARN, ERROR).
- **Body:** The actual message or structured data.

```
// OTLP Log Record Example
logger.emit({
  body: "User login successful",
  severityNumber: SeverityNumber.INFO,
})
```

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OpenTelemetry Components

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Instrumenting Your App

The vanilla way (only traces)

tracing.js

```
import { defaultResource, resourceFromAttributes } from '@opentelemetry/resources';
import { BatchSpanProcessor, ConsoleSpanExporter } from '@opentelemetry/sdk-trace-base';
import { OTLPTraceExporter } from '@opentelemetry/exporter-trace-otlp-http';
import { WebTracerProvider } from '@opentelemetry/sdk-trace-web';
import { ZoneContextManager } from '@opentelemetry/context-zone';

// 1. Identify the service and set metadata (e.g.: `service.name`)
const resource = defaultResource().merge(resourceFromAttributes({
  'service.name': 'my-webapp'
}));

// 2. Initialize the Tracer Provider with the defined resource
const provider = new WebTracerProvider({
  resource,
  // 3. Configure a Span processor and exporter for data egress
  spanProcessors: [new BatchSpanProcessor(new OTLPTraceExporter({
    url: 'https://collector.example.com/v1/traces'
  }))]
});

// 4. Register the Provider (optionally with a specific context manager)
provider.register({ contextManager: new ZoneContextManager()});
```

Package Management

Needs tooling

Several APIs to learn

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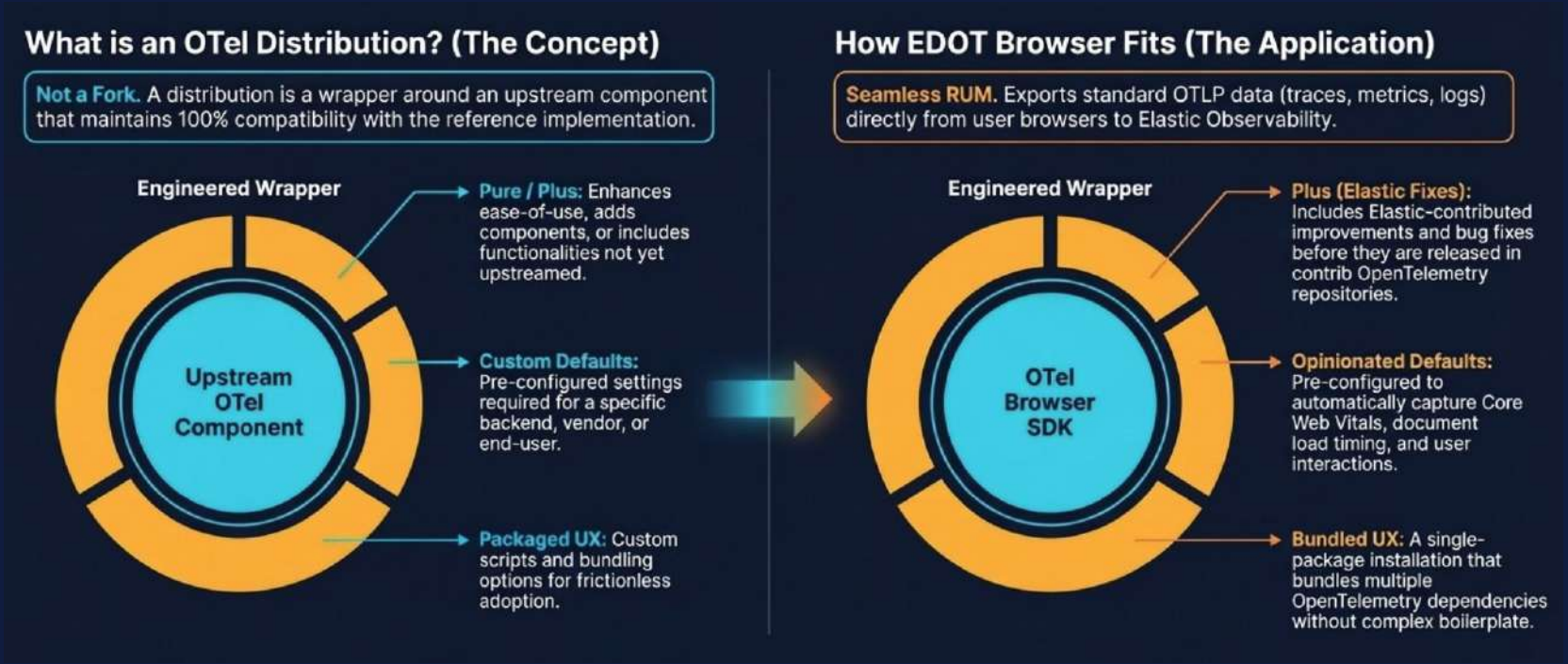
How to make it easier?



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Instrumenting Your App

EDOT Browser

tracing.js

```
import { startBrowserSdk } from '@elastic/opentelemetry-browser';

// Start passing the configuration options
startBrowserSdk({
  serviceName: 'my-webapp',
  otlpEndpoint: 'https://collector.example.com',
}));
```

Single Package

Tooling optional

Simplified API

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DEMO

OpenTelemetry Browser SIG: Phase 1 Summary

Core Principles

Developing an OpenTelemetry implementation highly optimized and tailored for browser environments. Prioritizing browser-specific instrumentations.

Key Objectives

Establish a dedicated browser repository. Optimize package size and dependencies for mobile and resource-constrained environments.

Initial Instrumentation Focus

Capture vital performance metrics: Browser Navigation, Web Exceptions, User Actions, Navigation Timing, and Core Web Vitals.

Browser Fundamentals

Reviewing Tracing, Metrics, and Logs APIs for browser-specific data modeling needs like Sessions, Resources, and Navigation.

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Useful links

Core Repository

The primary SDK for Node.js and Browser tracing, metrics, and logs.

`github.com/open-telemetry/opentelemetry-js`

Contrib Repository

Community-maintained instrumentations and resource detectors.

`github.com/open-telemetry/opentelemetry-js-contrib`

Browser Repository

Instrumentations and other components focused on browser-specific optimizations.

`github.com/open-telemetry/opentelemetry-browser`

Website & Docs

Official documentation and getting started guides for JS.

`opentelemetry.io/docs/languages/js/`

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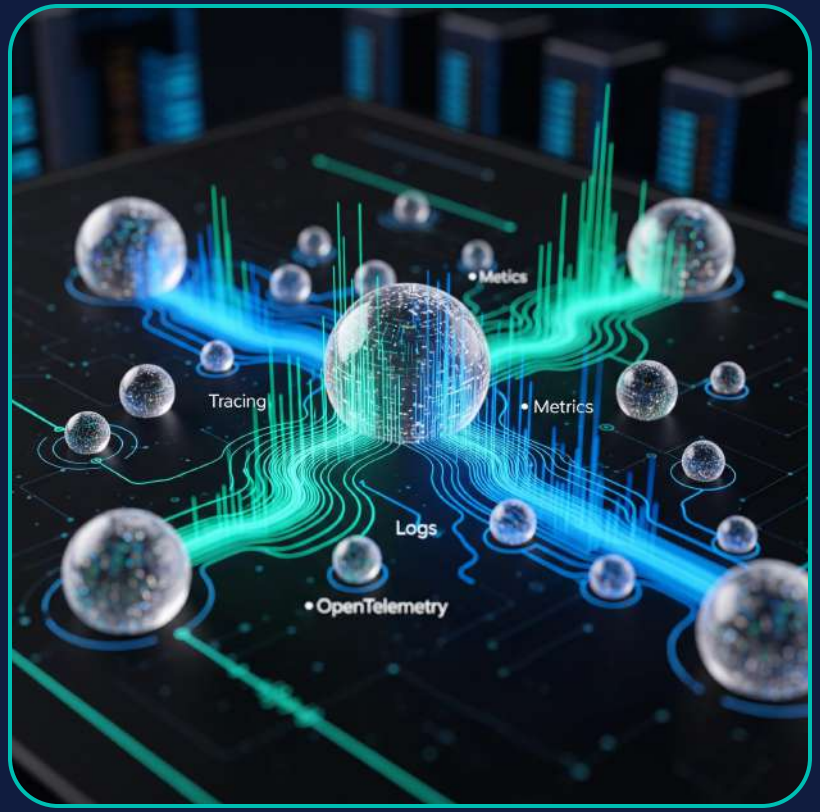
Thank You!

Questions & Discussion



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